

ATMA RAM SANATAN DHARMA COLLEGE

(UNIVERSITY OF DELHI)

Database Management System

(Practical File)

SUBMITTED BY

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Practical – I

- I. Create and use the following student-society database schema for a college to answer the given (sample) queries using the standalone SQL editor.

STUDENT	<u>Roll No</u>	StudentName	Course	DOB
	Char(6)	Varchar(20)	Varchar(10)	Date

SOCIETY	<u>SocID</u>	SocName	MentorName	TotalSeats
	Char(6)	Varchar(20)	Varchar(15)	Unsigned int

ENROLLMENT	<u>Roll No</u>	<u>SID</u>	DateOfEnrollment
	Char(6)	Char(6)	Date

Here Rollno (ENROLLMENT) and SID (ENROLLMENT) are foreign keys.

```
CREATE TABLE STUDENT (  
    RollNo CHAR(6) PRIMARY KEY,  
    StudentName VARCHAR(20),  
    Course VARCHAR(10),  
    DOB DATE  
);  
CREATE TABLE SOCIETY (  
    SocID CHAR(6) PRIMARY KEY,  
    SocName VARCHAR(20),  
    MentorName VARCHAR(15),  
    TotalSeats UNSIGNED INT  
);  
CREATE TABLE ENROLLMENT (  
    RollNo CHAR(6),  
    SID CHAR(6),  
    DateOfEnrollment DATE,  
    PRIMARY KEY (RollNo, SID),  
    FOREIGN KEY (RollNo) REFERENCES STUDENT(RollNo),  
    FOREIGN KEY (SID) REFERENCES SOCIETY(SocID)  
);
```

```
INSERT INTO STUDENT VALUES
```

```
('STU001', 'Arjun Mehta', 'Computer Science', '2002-05-12'),  
( 'STU002', 'Priya Singh', 'Chemistry', '2001-11-20'),  
( 'STU003', 'Ravi Kumar', 'Economics', '2003-03-05'),  
( 'STU004', 'Anita Gupta', 'Computer Science', '2000-07-15'),  
( 'XSTU009', 'Xavier Roy', 'Physics', '2001-01-10'),  
( 'ZSTU019', 'Zara Khan', 'Chemistry', '2004-02-22');
```

```
INSERT INTO SOCIETY VALUES
```

```
('S1', 'Debating', 'Prof. Gupta', 20),  
( 'S2', 'Dancing', 'Dr. Sharma', 25),  
( 'S3', 'Sashakt', 'Prof. Rao', 30),  
( 'S4', 'NSS', 'Dr. Mehta', 40);
```

```
INSERT INTO ENROLLMENT VALUES
```

```
('STU001', 'S1', '2024-07-01'),  
( 'STU001', 'S2', '2024-07-02'),  
( 'STU002', 'S3', '2024-07-03'),  
( 'STU003', 'S1', '2024-07-04'),  
( 'STU004', 'S4', '2024-07-05'),
```

1. Retrieve names of students enrolled in any society.

```
SELECT DISTINCT S.StudentName
```

```
FROM STUDENT S
```

```
JOIN ENROLLMENT E ON S.RollNo = E.RollNo;
```

StudentName
Arjun Mehta
Priya Singh
Ravi Kumar
Anita Gupta
Xavier Roy
Zara Khan

2. Retrieve all society names.

```
SELECT SocName FROM SOCIETY;
```

SocName
Debating
Dancing
Sashakt
NSS

3. Retrieve students' names starting with the letter 'A'.

```
SELECT StudentName FROM STUDENT  
WHERE StudentName LIKE 'A%';
```

StudentName
Arjun Mehta
Anita Gupta

4. Retrieve students' details studying in courses 'computer science' or 'chemistry'.

```
SELECT * FROM STUDENT  
WHERE Course IN ('Computer Science', 'Chemistry');
```

RollNo	StudentName	Course	DOB
STU001	Arjun Mehta	Computer Science	2002-05-12
STU002	Priya Singh	Chemistry	2001-11-20
STU004	Anita Gupta	Computer Science	2000-07-15
ZSTU019	Zara Khan	Chemistry	2004-02-22

5. Retrieve students' names whose roll no either starts with 'X' or 'Z' and ends with '9'

```
SELECT StudentName FROM STUDENT  
WHERE (RollNo LIKE 'X%9' OR RollNo LIKE 'Z%9');
```

StudentName
Xavier Roy
Zara Khan

6. Find society details with more than N TotalSeats where N is to be input by the user

```
SELECT * FROM SOCIETY  
WHERE TotalSeats > 33
```

SocID	SocName	MentorName	TotalSeats
S4	NSS	Dr. Mehta	40

7. Update society table for the mentor name of a specific society

```
UPDATE SOCIETY  
SET MentorName = 'New Mentor'  
WHERE SocID = 'S1';
```


8. Find society names in which more than five students have enrolled

```
SELECT So.SocName
FROM SOCIETY So
JOIN ENROLLMENT E ON So.SocID = E.SID
GROUP BY So.SocID, So.SocName
HAVING COUNT(E.RollNo) > 5;
```

9. Find the name of the youngest student enrolled in society 'NSS'

```
SELECT S.StudentName
FROM STUDENT S
JOIN ENROLLMENT E ON S.RollNo = E.RollNo
JOIN SOCIETY So ON E.SID = So.SocID
WHERE So.SocName = 'NSS'
ORDER BY S.DOB DESC
LIMIT 1;
```

StudentName

Anita Gupta

10. Find the name of the most popular society (on the basis of enrolled students)

```
SELECT So.SocName
FROM SOCIETY So
JOIN ENROLLMENT E ON So.SocID = E.SID
GROUP BY So.SocID, So.SocName
ORDER BY COUNT(E.RollNo) DESC
LIMIT 1;
```

SocName

Debating

11. Find the name of two least popular societies (on the basis of enrolled students)

```
SELECT So.SocName
FROM SOCIETY So
LEFT JOIN ENROLLMENT E ON So.SocID = E.SID
GROUP BY So.SocID, So.SocName
ORDER BY COUNT(E.RollNo) ASC
LIMIT 2;
```

SocName
NSS
Dancing

12. Find the students names who are not enrolled in any society.

```
SELECT S.StudentName
FROM STUDENT S
LEFT JOIN ENROLLMENT E ON S.RollNo = E.RollNo
WHERE E.SID IS NULL;
```

13. Find the students names enrolled in at least two societies

```
SELECT S.StudentName
FROM STUDENT S
JOIN ENROLLMENT E ON S.RollNo = E.RollNo
GROUP BY S.RollNo, S.StudentName
HAVING COUNT(DISTINCT E.SID) >= 2;
```

StudentName
Arjun Mehta
Zara Khan

14. Find society names in which maximum students are enrolled

```
SELECT So.SocName
FROM SOCIETY So
JOIN ENROLLMENT E ON So.SocID = E.SID
GROUP BY So.SocID, So.SocName
ORDER BY COUNT(E.RollNo) DESC
LIMIT 1;
```

SocName
Debating

15. Find names of all students who have enrolled in any society and society names in which at least one student has enrolled

```
SELECT DISTINCT S.StudentName, So.SocName
FROM STUDENT S
JOIN ENROLLMENT E ON S.RollNo = E.RollNo
JOIN SOCIETY So ON E.SID = So.SocID;
```

StudentName	SocName
Arjun Mehta	Debating
Arjun Mehta	Dancing
Priya Singh	Sashakt
Ravi Kumar	Debating
Anita Gupta	NSS
Xavier Roy	Debating
Zara Khan	Dancing
Zara Khan	Sashakt

16. Find names of students who are enrolled in any of the three societies 'Debating', 'Dancing' and 'Sashakt'.

```
SELECT DISTINCT S.StudentName
FROM STUDENT S
JOIN ENROLLMENT E ON S.RollNo = E.RollNo
JOIN SOCIETY So ON E.SID = So.SocID
WHERE So.SocName IN ('Debating','Dancing','Sashakt');
```

StudentName
Arjun Mehta
Priya Singh
Ravi Kumar
Xavier Roy
Zara Khan

17. Find society names such that its mentor has a name with 'Gupta' in it.

```
SELECT SocName FROM SOCIETY
WHERE MentorName LIKE '%Gupta%';
```

18. Find the society names in which the number of enrolled students is only 10% of its capacity.

```
SELECT So.SocName
FROM SOCIETY So
JOIN ENROLLMENT E ON So.SocID = E.SID
GROUP BY So.SocID, So.SocName, So.TotalSeats
HAVING COUNT(E.RollNo) = 0.1 * So.TotalSeats;
```

19. Display the vacant seats for each society.

```
SELECT So.SocName, So.TotalSeats - COUNT(E.RollNo) AS VacantSeats
FROM SOCIETY So
LEFT JOIN ENROLLMENT E ON So.SocID = E.SID
GROUP BY So.SocID, So.SocName, So.TotalSeats;
```

SocName	VacantSeats
Debating	17
Dancing	23
Sashakt	28
NSS	39

20. Increment Total Seats of each society by 10%

```
UPDATE SOCIETY
SET TotalSeats = ROUND(TotalSeats * 1.1);
```

21. Add the enrollment fees paid ('yes'/'No') field in the enrollment table.

```
ALTER TABLE ENROLLMENT  
ADD COLUMN FeesPaid ENUM('Yes','No') DEFAULT 'No';
```

22. Update date of enrollment of society id 's1' to '2018-01-15', 's2' to the current date and 's3' to '2018-01-02'.

```
UPDATE ENROLLMENT SET DateOfEnrollment = '2018-01-15' WHERE SID='S1';  
UPDATE ENROLLMENT SET DateOfEnrollment = CURRENT_DATE WHERE SID='S2';  
UPDATE ENROLLMENT SET DateOfEnrollment = '2018-01-02' WHERE SID='S3';
```

23. Create a view to keep track of society names with the total number of students enrolled in it.

```
CREATE VIEW SocietyEnrollment AS  
SELECT So.SocName, COUNT(E.RollNo) AS TotalEnrolled  
FROM SOCIETY So  
LEFT JOIN ENROLLMENT E ON So.SocID = E.SID  
GROUP BY So.SocID, So.SocName;
```

24. Find student names enrolled in all the societies.

```
SELECT COUNT(*)  
FROM SOCIETY So  
JOIN ENROLLMENT E ON So.SocID = E.SID  
GROUP BY So.SocID  
HAVING COUNT(E.RollNo) > 5;
```

25. Count the number of societies with more than 5 students enrolled in it

```
SELECT COUNT(*)  
FROM SOCIETY So  
JOIN ENROLLMENT E ON So.SocID = E.SID  
GROUP BY So.SocID  
HAVING COUNT(E.RollNo) > 5;
```

26. Add column Mobile number in student table with default value '9999999999'

```
ALTER TABLE STUDENT  
ADD COLUMN Mobile VARCHAR(10) DEFAULT '9999999999';
```

27. Find the total number of students whose age is > 20 years.

```
SELECT COUNT(*)  
FROM STUDENT  
WHERE TIMESTAMPDIFF(YEAR, DOB, CURDATE()) > 20;
```

COUNT(*)
6

28. Find names of students who were born in 2001 and are enrolled in at least one society.

```
FROM STUDENT S  
JOIN ENROLLMENT E ON S.RollNo = E.RollNo  
WHERE YEAR(S.DOB) = 2001;
```

StudentName
Priya Singh
Xavier Roy

29. Count all societies whose name starts with 'S' and ends with 't' and at least 5 students are enrolled in the society.

```
SELECT COUNT(*)  
FROM SOCIETY So  
JOIN ENROLLMENT E ON So.SocID = E.SID  
WHERE So.SocName LIKE 'S%t'  
GROUP BY So.SocID  
HAVING COUNT(E.RollNo) >= 5;
```

30. Display the following information: Society name Mentor name Total Capacity Total Enrolled Unfilled Seats

```
SELECT So.SocName, So.MentorName, So.TotalSeats,  
COUNT(E.RollNo) AS TotalEnrolled,  
So.TotalSeats - COUNT(E.RollNo) AS UnfilledSeats
```

SocName	MentorName	TotalCapacity	TotalEnrolled	UnfilledSeats
Debating	New Mentor	22	3	19
Dancing	Dr. Sharma	28	2	26
Sashakt	Prof. Rao	33	2	31
NSS	Dr. Mehta	44	1	43

Practical – II

Do the following database administration commands: Create user, create role, grant privileges to a role, revoke privileges from a role, create index

```
CREATE USER 'student_user'@'localhost' IDENTIFIED BY 'password123';  
CREATE ROLE 'society_role';  
GRANT SELECT, INSERT ON college.STUDENT TO 'society_role';  
GRANT 'society_role' TO 'student_user'@'localhost';  
REVOKE INSERT ON college.STUDENT FROM 'society_role';  
CREATE INDEX idx_student_name ON STUDENT(StudentName);
```